SECTION A

*Write about a quarter of a page each on any **four** of the following topics. Clearly state if you agree or disagree with each statement.*

Deduct marks for each major thing missed, and for each wrong statement. In general, be relatively generous if the answer makes sense and contains the main ideas.

2. *MAE, RMSE, MAPE and MASE are all similar measures; hence it does not matter which one we use.*

- This statement is not true, as these are very different measures with different properties. 1
- The first two measures are scale dependent and if the series are measured on different scales these cannot be used for comparison across these series. The latter two measures are scale independent 1
- With MAPE we need to be careful as it is not specified for series with zeros and it will be very skewed if we have observations close to zero. 1
- MASE is robust to this as the scaling factor comes from the MAE over the training set of a simple method such as the naive for non-seasonal time series. 2

3. *Always choose the model with the smallest sum of squared errors.*

- This statement is not true, as simply considering sum of squared errors is problematic, especially for the purpose of forecasting. 1
- Selecting a model based on SSE is problematic because models can be built to over-fit the data which has an adverse effect on forecasting. 1
- For example, in a regression model adding another variable will always reduce the sum of squared errors, even if the additional variable is of no value in forecasting. We can always make an ARIMA model more complicated to fit the data better. 1
- Instead, it is best to choose models based on information criteria such as the AICc. Such criteria penalise the sum of squared errors to take into account the number of parameters in the model / to take into account the complexity of the model. 2

SECTION B

1. *Using Figures 1–3, describe the weekly mortality rate for Australia. Carefully comment on the interesting features of all three plots. [6 marks]*

- Time plot shows annual seasonality (with peaks during the winter weeks) 1
- Time plot shows an increasing trend over the last couple of years - covid effect 1
- Time plot shows a higher peak compared to other pre-Covid years for 2017; a different shape (fewer deaths over the winter period) for the winter peak of 2020; and an unusual summer peak in 2022. 1
- Season plot verifies the annual seasonality, with peaks during the weeks of June to September, the increasing trend with 2021 and 2022 being higher than the other years, the unusually high peak for 2017 compared to the other pre-Covid years, the lower rate for the winter weeks for 2020; and the unusual summer peak of 2022. 1
- STL decomposition verifies the features we have already seen in the trend panel, both in terms of an unusual bump in trend in 2017, the lower trend in the winter of 2020 and the increasing trend in the last few years 1
- The remainder shows a few outliers with the most pronounced at the beginning of 2022 1

2. *For the STL decomposition shown in Figure 3, discuss the effect of the window sizes chosen for the trend and seasonal components. How would the results have changed with smaller or larger values chosen in each case? [4 marks]*

- The trend window includes 9 weeks. This is quite small. As a result the trend is not very smooth and possibly contains outliers such as the 2022 bump. Increasing this would make the trend smoother and possibly push to the remainder some possible outliers. $\boxed{2}$

- The seasonal window is set to 'periodic' and as a result the seasonal component is not allowed to change over time. A smaller window would allow the seasonal component to vary but we only have 8 years of data so the window cannot vary by much . $\boxed{2}$

3.(GA: I think we need to be lenient here if they pick up/comment on short v long run forecasts. We should keep this in mind)

(a) *Seasonal naïve method using annual seasonality.*

- Suitable. It would generate the annual seasonality. Not ok if the trend continues. $\boxed{1}$

(b) *Seasonal naïve method using weekly seasonality.*

- Not suitable. There is no weekly seasonality. $\boxed{1}$

(c) *An STL decomposition with an ETS to forecast the seasonally adjusted component, and seasonal naïve for the seasonal component.*

- Suitable. This would work well as the possible trend will also be projected. $\boxed{1}$

(d) *Holt's method with damped trend.*

- Not suitable. It would not allow for the annual seasonality. $\boxed{1}$

(e) *ETS(A,N,A).*

- Not suitable. It would not allow for the annual seasonality. $\boxed{1}$

(f) *ETS(M,A,M) with annual seasonality.*

- Not suitable. There are too many parameters to estimate for annual seasonality with weekly data. $\boxed{1}$

Lesson/Reminder

- ETS models are not suitable for modelling annual seasonality with weekly data (52 weeks).
- For daily data they capture well weekly seasonality (7-days in a week) but again not annual (365 days). Short term forecasting a few days ahead should be fine.

SECTION C

1. *Why can't you fit an ETS model to this data set? [3 marks]*

Too many parameters ($52 - 1$ for the weeks in a year) will need to be estimated, in order to estimate the seasonal component of the model. $\boxed{3}$

2. *You decide to use the STL decomposition shown in Figure 3, with ETS used for the seasonally adjusted data. Write down the equations for the fitted model, including the STL equation, and the equations for the components. [6 marks]*

(I am using the notation we have in the book for forecasting with decomposition)

STL decomposition

$$y_t = \hat{S}_t + \hat{A}_t$$

$\boxed{2}$

Seasonal naïve

$$\hat{S}_t = \hat{S}_{t-52} + \eta_t$$

$\boxed{2}$

ETS(M,N,N) on the seasonally adjusted component

$$\hat{A}_t = \ell_{t-1}(1 + \varepsilon_t)$$
$$\ell_t = \ell_{t-1}(1 + \alpha\varepsilon_t)$$

$\boxed{2}$

3. *The residuals for the model are shown in Figure 4.*

What is causing the outlier at the start of 2022? [2 marks]

The unusual increase in the mortality rate at the beginning of 2022 which has been passed into the irregular component of the STL decomposition and hence in the residuals of the model. This is due to Covid (and in particular due to relaxing restrictions in Australia). $\boxed{2}$

Why are there no residuals for 2015? [2 marks]

There are no residuals as the seasonal naive method has been implemented for the seasonal component, and it needs at least a year of data to generate a forecast. $\boxed{2}$

Do you think the residuals would pass a white noise test? [2 marks]

The residuals look like white noise (no obvious patterns or autocorrelation left over) so yes I think they would pass the test. $\boxed{2}$

4. *The forecasts for the model are shown in Figure 5.*

What features of the data have not been captured by the model, and how does this affect the resulting forecasts? [3 marks]

The main strong feature not captured (and projected by the model) is the trend of the last couple of years. The Seasonally Adjusted component was projected with an ETS(M,N,N) hence no trend. Also the spike at the beginning of 2022 has not been captured and appears in the residuals. The model not capturing these results leads to wider prediction intervals. 1 1 1

Discuss how the large prediction intervals would affect how this model would be used to inform policy decisions around mortality risks? [2 marks]

The large prediction intervals highlight/reflect high uncertainty. This should clearly be communicated to policy makers when presenting them with the forecasts. 2

[Total: 20 marks]

— END OF SECTION C —